

Songlin Yang

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Education

Massachusetts Institute of Technology , Cambridge, MA, USA Ph.D. in Computer Science Advisor: Yoon Kim	2023-2025
ShanghaiTech University , Shanghai, China M.S. in Computer Science Advisor: Kewei Tu	2020-2023
Southern University of Science and Technology , Shenzhen, China B.Eng in Computer Science	2016-2020

Employment

Thinking Machines Lab , San Francisco, CA, USA <i>Technical Lead, Pretraining Science Team (Full-Time)</i>	Jan. 2026 – Present
Thinking Machines Lab , San Francisco, CA, USA <i>Part-Time</i>	Sep. 2025 – Dec. 2025
MIT-IBM Watson AI Lab , Cambridge, MA, USA <i>Research Intern</i> Mentors: Rameswar Panda, Yoon Kim	Jun. 2025 – Aug. 2025
NVIDIA Research , Santa Clara, CA, USA <i>Research Intern</i> Mentors: Ali Hatamizadeh, Jan Kautz	May 2024 – Aug. 2024

Teaching

Teaching Assistant at Massachusetts Institute of Technology Natural Language Processing Instructor: Yoon Kim and Jacob Andreas	Fall 2025
Teaching Assistant at ShanghaiTech University Natural Language Processing Instructor: Kewei Tu	Spring 2022
Teaching Assistant at ShanghaiTech University Artificial Intelligence Instructor: Kewei Tu	Fall 2021
Teaching Assistant at Southern University of Science and Technology Artificial Intelligence Instructor: Ke Tang	Fall 2019

Referred Conference Publications

- [1] Yanhong Li*, **Songlin Yang***, Shawn Tan, Mayank Mishra, Rameswar Panda, Jiawei Zhou, and Yoon Kim. *Distilling to Hybrid Attention Models via KL-Guided Layer Selection*. In: *The Fourteenth International Conference on Learning Representations (ICLR)*. 2026.
- [2] **Songlin Yang**, Yikang Shen, Kaiyue Wen, Shawn Tan, Mayank Mishra, Liliang Ren, Rameswar Panda, and Yoon Kim. *PaTH Attention: Position Encoding via Accumulating Householder Transformations*. In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems*. 2025.
- [3] Xingyang Li, Muyang Li, Tianle Cai, Haocheng Xi, Shuo Yang, Yujun Lin, Lvmin Zhang, **Songlin Yang**, Jinbo Hu, Kelly Peng, Maneesh Agrawala, Ion Stoica, Kurt Keutzer, and Song Han. *Radial Attention: $O(n \log n)$ Sparse Attention for Long Video Generation*. In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS)*. 2025.
- [4] Zihan Qiu, Zekun Wang, Bo Zheng, Zeyu Huang, Kaiyue Wen, **Songlin Yang**, Rui Men, Le Yu, Fei Huang, Suozhi Huang, Dayiheng Liu, Jingren Zhou, and Junyang Lin. *Gated Attention for Large Language Models: Non-linearity, Sparsity, and Attention-Sink-Free*. In: *The Thirty-ninth Annual Conference on Neural Information Processing Systems (NeurIPS)*. 2025.
- [5] Yi Lu, Jing Nathan Yan, **Songlin Yang**, Justin T Chiu, Siyu Ren, Fei Yuan, Wenting Zhao, Zhiyong Wu, and Alexander M Rush. *A Controlled Study on Long Context Extension and Generalization in LLMs*. In: *Second Conference on Language Modeling (COLM)*. 2025.
- [6] **Songlin Yang**, Jan Kautz, and Ali Hatamizadeh. *Gated Delta Networks: Improving Mamba2 with Delta Rule*. In: *The Thirteenth International Conference on Learning Representations (ICLR)*. 2025.
- [7] Shawn Tan, **Songlin Yang**, Aaron Courville, Rameswar Panda, and Yikang Shen. *Scaling Stick-Breaking Attention: An Efficient Implementation and In-depth Study*. In: *The Thirteenth International Conference on Learning Representations (ICLR)*. 2025.
- [8] **Songlin Yang**, Bailin Wang, Yu Zhang, Yikang Shen, and Yoon Kim. *Parallelizing Linear Transformers with the Delta Rule over Sequence Length*. In: *Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS)*. 2024.
- [9] Yu Zhang*, **Songlin Yang***, Ruijie Zhu, Yue Zhang, Leyang Cui, Yiqiao Wang, Bolun Wang, Freda Shi, Bailin Wang, Wei Bi, Peng Zhou, and Guohong Fu. *Gated Slot Attention for Efficient Linear-Time Sequence Modeling*. In: *Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS)*. 2024.
- [10] Zhen Qin*, **Songlin Yang***, Weixuan Sun, Xuyang Shen, Dong Li, Weigao Sun, and Yiran Zhong. *HGRN2: Gated Linear RNNs with State Expansion*. In: *First Conference on Language Modeling (COLM)*. 2024.
- [11] **Songlin Yang***, Bailin Wang*, Yikang Shen, Rameswar Panda, and Yoon Kim. *Gated Linear Attention Transformers with Hardware-Efficient Training*. In: *Forty-first International Conference on Machine Learning (ICML)*. 2024.
- [12] Zhen Qin*, **Songlin Yang***, and Yiran Zhong. *Hierarchically Gated Recurrent Neural Network for Sequence Modeling*. In: *Thirty-seventh Conference on Neural Information Processing Systems (NeurIPS)*. 2023.
- [13] Wei Liu*, **Songlin Yang***, Yoon Kim, and Kewei Tu. *Simple Hardware-Efficient PCFGs with Independent Left and Right Productions*. In: *Findings of the Association for Computational Linguistics (EMNLP Findings)*. 2023.
- [14] Pengyu Ji, **Songlin Yang**, and Kewei Tu. *Improving Span Representation by Efficient Span-Level Attention*. In: *Findings of the Association for Computational Linguistics (EMNLP Findings)*. 2023.
- [15] Zhaohui Yan, **Songlin Yang**, Wei Liu, and Kewei Tu. *Joint Entity and Relation Extraction with Span Pruning and Hypergraph Neural Networks*. In: *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2023.
- [16] **Songlin Yang**, Roger Levy, and Yoon Kim. *Unsupervised Discontinuous Constituency Parsing with Mildly Context-Sensitive Grammars*. In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*. 2023.

- [17] **Songlin Yang** and Kewei Tu. *Don't Parse, Choose Spans! Continuous and Discontinuous Constituency Parsing via Autoregressive Span Selection*. In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*. 2023.
- [18] Wei Liu, **Songlin Yang**, and Kewei Tu. *Structured Mean-Field Variational Inference for Higher-Order Span-Based Semantic Role Labeling*. In: *Findings of the Association for Computational Linguistics (ACL Findings)*. 2023.
- [19] **Songlin Yang** and Kewei Tu. *Headed-Span-Based Projective Dependency Parsing*. In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2022.
- [20] **Songlin Yang** and Kewei Tu. *Bottom-Up Constituency Parsing and Nested Named Entity Recognition with Pointer Networks*. In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2022.
- [21] Chao Lou, **Songlin Yang**, and Kewei Tu. *Nested Named Entity Recognition as Latent Lexicalized Constituency Parsing*. In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*. 2022.
- [22] **Songlin Yang***, Wei Liu*, and Kewei Tu. *Dynamic Programming in Rank Space: Scaling Structured Inference with Low-Rank HMMs and PCFGs*. In: *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*. 2022.
- [23] **Songlin Yang** and Kewei Tu. *Combining (Second-Order) Graph-Based and Headed-Span-Based Projective Dependency Parsing*. In: *Findings of the Association for Computational Linguistics (ACL Findings)*. 2022.
- [24] **Songlin Yang** and Kewei Tu. *Semantic Dependency Parsing with Edge GNNs*. In: *Findings of the Association for Computational Linguistics (EMNLP Findings)*. 2022.
- [25] **Songlin Yang**, Yanpeng Zhao, and Kewei Tu. *PCFGs Can Do Better: Inducing Probabilistic Context-Free Grammars with Many Symbols*. In: *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*. 2021.
- [26] **Songlin Yang**, Yanpeng Zhao, and Kewei Tu. *Neural Bi-Lexicalized PCFG Induction*. In: *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (ACL)*. 2021.
- [27] **Songlin Yang**, Yong Jiang, Wenjuan Han, and Kewei Tu. *Second-Order Unsupervised Neural Dependency Parsing*. In: *Proceedings of the 28th International Conference on Computational Linguistics (COLING)*. 2020.

Invited Talks

- [28] **Songlin Yang**. *Linear Transformers for Efficient Sequence Modeling*. Invited talk, Stanford University. Aug. 2024.
- [29] **Songlin Yang**. *Gated Linear Recurrence for Efficient Sequence Modeling*. Invited talk, LinkedIn Corporation. June 2024.
- [30] **Songlin Yang**. *Gated Linear Recurrence for Efficient Sequence Modeling*. Invited talk, Cornell Tech. Apr. 2024.

Open-Sourced Project

1. Flash Linear Attention

A Triton-Based Library for Hardware-Efficient Implementations of Linear Attention Mechanism.
<https://github.com/sustcsonglin/flash-linear-attention/>